CAROLINA MONZÓ, PhD

Data Analyst | Computational Biologist | Bioinformatician

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PROFESSIONAL SUMMARY

Bioinformatician and Data Analyst with over 9 years of experience in cross-sectional and longitudinal phenotypic and omic data analysis. Expert in developing bioinformatics software tools using Python and R, with deep expertise in transcriptomics, single-cell analysis, and long-read sequencing. Proven track record in wet-lab experimental design, statistical modeling, and leadership in academic and research settings. Marie Skłodowska-Curie Fellow and Principal Investigator on competitive grants.

EXPERIENCE

Marie Skłodowska-Curie Actions Postdoctoral Fellow 2024 – Present
Institute for Integrative Systems Biology (I2SysBio), Valencia, Spain

- Design and develop tools for long-read transcriptomics data analysis (lrRNA-Seq, machine learning).
- Supervise PhD and MSc students, mentoring them in computational biology workflows.
- Design independent studies and secure competitive funding (PI on Oxford Nanopore and JAE Intro grants).
- Serve as a board member in I2SysBio and co-lead the Junior Hub on Bioinformatics and Computational Biology (CSIC).

Postdoctoral Researcher 2022 – 2024
Institute for Integrative Systems Biology (I2SysBio), Valencia, Spain

- Developed MOSim, a Bioconductor package for simulating bulk and single-cell multi-omics data (RNA-Seq, ATAC-Seq, Methyl-Seq etc).
- Supervised MSc students and participated in funding applications.
- Organized seminars and managed scientific communication channels.

Ph.D. Researcher / Doctoral Candidate 2019 – 2022
Max Planck Institute for Biology of Ageing, Cologne, Germany

- Analyzed multi-omics datasets (RNA-Seq, 16S microbiome, AIRR-Seq, proteomics, metabolomics) to study aging interventions.
- Performed survival analysis and applied machine learning models to longitudinal mouse data.
- Conducted wet-lab experiments including nucleic acid extraction and library preparation.

Senior Technician 2017 – 2018
Health Research Institute (INCLIVA), Valencia, Spain

- Managed clinical genomics workflows and supported hematology diagnostics.

Bioinformatics Technician 2017
Health in Code, Valencia, Spain

- Developed pipelines for variant calling and clinical reporting.

EDUCATION

Ph.D. in Natural Sciences (Cum Laude) 2022
Universität zu Köln / Max Planck Institute for Biology of Ageing, Germany
Thesis: The effect of Dietary Restriction on the Microbiome and the Adaptive Immune System.

M.Sc. in Bioinformatics (Outstanding) 2018
Universitat de València, Spain
Postgraduate Diploma in Medical Genetics 2015
Universitat de València, Spain
B.Sc. in Biology 2014
Universitat de València, Spain

SKILLS

Programming & Development: Python, R, Bash, Git, Bioconductor package development.
Data Analysis: Transcriptomics (Bulk/Single-cell/Long-reads), Genomics, Proteomics, Metabolomics, Longitudinal Statistics, Survival analysis, Machine Learning.
Infrastructure: Linux/Unix, High-Performance Computing (HPC), AWS, Docker/Singularity.
Laboratory: Library preparation (Illumina/ONT), Nucleic acid extraction, PCR.
Languages: English (Fluent), Spanish and Catalan (Native).

GRANTS & FUNDING

Principal Investigator, JAE Intro 2025 (CSIC) 2025 – 2026
Project: Software for differentiating technical noise vs. transcript divergency in long-reads.
Principal Investigator, Oxford Nanopore Grant 2025 – 2026
Project: Investigating RNA degradation patterns using Nanopore sequencing.
Principal Investigator, Marie Skłodowska-Curie Postdoctoral Fellowship 2024 – 2026
Project: Deciphering RNA biological noise in aging.

SELECTED PUBLICATIONS

- **Nature Reviews Genetics (2025)**: Review on long-read transcriptomics approaches (First/Corresponding Author).
- **Genome Research (2025)**: Perspective on long-read transcriptomics challenges (First Author).
- **Genome Research (2025)**: Quality assessment of long read data in multisample lrRNA-seq experiments with SQANTI-reads (Co-author).
- **Briefings in Bioinformatics (2025)**: MOSim: Multi-omics simulation for bulk and single-cell data (First Author).
- **Nature Aging (2025)**: Trametinib and rapamycin combine to extend healthspan (Co-author).
- **Cell Reports (2023)**: Dietary restriction mitigates age-associated B cell receptor repertoire decline (First Author).

AWARDS

1st Prize, Best Highlight Talk (XV JBI Symposium, 2025)
2nd Prize, Best Flash Talk (SEH2Bioinfo, 2025)
Finalist, Best PhD Thesis (Spanish Society for Bioinformatics/Computational Biology, 2024)
1st Prize, Best Poster (MOSim, SEBiBC Congress, 2024)

LEADERSHIP & MENTORING

Supervision: 1 PhD student, 14 MSc students.
Leadership: Co-lead of Junior BCBHub-CSIC; Board Member at I2SysBio.
Teaching: Organizer/Lecturer for "Long-read Transcriptomics Summer School 2025", "Long-read transcriptomics workflows and applications 2024" and "CSIC4CSIC 2024" Workshop.
Mentoring: Organizer of "Adopt a Bioinformatician" mentoring program

For complete details, publications, courses, and interactive content:
https://carolinamonzo.github.io/